

# Andrew Liu

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## EDUCATION

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University of California, Santa Barbara  
College of Engineering - B.S. Electrical Engineering  
Expected Graduation: June 2028  
Cumulative GPA: 3.98/4.00

### Relevant Coursework:

Circuits and Systems (Analog and Digital) I-III + Laboratories I-III  
Fundamentals of Logic Design, Digital Design Principles, Digital Image Processing  
Physics: Electricity and Magnetism, Optics and Photonics, Modern Physics + Laboratory  
Computer Science: Data Structures and Algorithms, Object-Oriented Programming (C++)

## WORK EXPERIENCE

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### Electrical Engineering Intern | Hoplynk June 2026 - Present

- Designed, debugged, and integrated Power Distribution & Battery Monitoring Printed Circuit Boards
- Designed and integrated power distribution systems using COTS electronics under time constraints
- Created write-ups and documents to explain my design choices, allowing for smooth design reviews
- Assembled and tested various electronics using a DMM and running test software (e.g., radios)

### Avionics/Electrical Team Lead | Gaucho Rocket Project January 2026 - Present

- Design data acquisition systems and boards to log and stream rocket telemetry during flight
- Verify electronic components and sensor functionality with lab test equipment (DMM, Oscilloscope)
- Document test results and create protocols so the team has reference data and test instructions

## PROJECTS

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### Hoplynk Power Distribution & Battery Monitoring PCB (PCB Design) June 2026 - September 2026

- Designed a power distribution & battery charging board with DC inputs (12V-32V) + smart charger
- Used MOSFET transistors at IC pins to ensure pack level isolation during battery discharge & charge
- Optimized board size while also derating all components to below 50% of max voltage for reliability
- Ensured EMI from high-power switching areas did not affect analog readings and digital ICs
- Implemented digital switch to allow cut-off of a heavy load, increasing system runtime by ~86%
- Wrote and flashed STM32 firmware and python scripts to send battery data and commands via USB-C

### Gaucho Rocket Project Sensor Logger Printed Circuit Board (PCB Design) April 2026 - May 2026

- Designed a 4-layer PCB in KiCad to log temperature, pressure, acceleration data during rocket flight
- Selected integrated circuits for size, power, sensor compatibility, and avionics requirements
- Validated and tested key components and sensors (e.g., thermocouple amplifier) on breakout boards

### Boost Converter Printed Circuit Board (KiCad, Soldering, Power Electronics) March 2026 - April 2026

- Designed and prototyped a boost converter circuit, achieving an efficiency of ~66% (5V to 12V)
- Implemented pulse width modulation control with closed-loop voltage feedback using Arduino/ATtiny
- Designed PCB layout, taking into account trace width calculations and component placement

## SKILLS

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**Technical Skills:** Analog and Digital Circuit Analysis, Circuit Prototyping and Debugging, KiCad, PCB Design & Rework, Power Electronics, LTspice, C++, Python, Oscilloscope, Multimeter, Function Generator